

Compact, Planar and Conformal Antenna Array Solution for Electronic Warfare Applications

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Abstract—This work presents the design and development of a compact, wideband, and conformal antenna and its array tailored for EW application in next-generation missile and unmanned aerial vehicle (UAV) platforms. A tapered slot (Vivaldi-based) antenna architecture is employed to achieve wideband performance across the frequency range of 3–18 GHz, supporting high-resolution angle-of-arrival (AOA) detection and precise direction finding. To accommodate modern aerodynamic constraints, the array is designed on thin, flexible substrates such as Rogers laminates, allowing seamless integration onto curved surfaces, including hemispherical domes and UAV bodies. The conformal array configuration ensures mechanical adaptability across diverse platforms while preserving electromagnetic performance.

I. INTRODUCTION

The Vivaldi antenna, originally proposed by Gibson [1], is a type of end-fire traveling wave antenna known for its use of printed circuit technology, which simplifies both its fabrication and structural reliability. Its broadband behavior and end-fire radiation characteristics make it suitable for a range of applications, including satellite communication, weather monitoring, and surveillance systems. Despite these advantages, a single Vivaldi antenna element typically offers limited gain, prompting the use of antenna arrays to achieve higher performance levels. Early implementations predominantly used planar array configurations. The operating frequency of the Vivaldi antenna is primarily influenced by the geometrical parameters of the radiating element—specifically, its length, width, and tapered profile—which directly affect its radiation efficiency [2], [3]. As a result, Vivaldi arrays have become popular for their broadband capabilities and ease of integration [4].

Several design enhancements, such as the inclusion of arc-shaped slots, uniform slot-edge structures, and dielectric lenses, have been explored to improve gain and reduce size. However, these modifications often introduce greater complexity and lead to larger physical dimensions [5], [6]. In recent years, conformal antenna arrays have emerged as a promising alternative due to their ability to reduce aerodynamic drag, minimize radar cross-section, improve angular coverage, and lower structural costs [3].

In this paper, the designs of a planar and conformal Vivaldi-like antenna and its array of 4 such elements are designed and discussed. It operates in a wide frequency range with

minimal form factor and achieves a consistent impedance match and gain over a 3–18 GHz range.

II. ANTENNA SELECTION

Ultra-wideband antennas exhibit a wide impedance bandwidth but not necessarily a stable radiation pattern, as required in most radar applications. Moreover, in addition to the large bandwidth and specific radiation pattern characteristics, the installation space environment usually impacts the antenna design and performance, as

it mainly constrains the operating frequency range. As a result, miniaturization techniques need to be implemented to save the installation space and to ensure an optimal trade-off between the antenna size and the radiation characteristics. For UWB radiating elements, this challenge is even harder to fulfill, as gain, beam width, and efficiency should be properly guaranteed. To this aim, several research activities, in both academic and industrial fields, are in progress for the identification of new wideband antenna geometries satisfying the above system requirements and constraints in an increasingly extended frequency band. In particular, they pertain to the planar antennas and they are focused on the improvement of both frequency band and gain uniformity, as well as on the polarization management capabilities, size, and weight reduction.

Wideband antennas like Vivaldi, spiral, sinuous, and LPDA are crucial for various applications due to their ability to operate effectively across a wide range of frequencies. Broad bandwidth is essential for high data rate communication, radar systems, and other applications requiring the transmission and reception of signals across a wide spectrum. UWB antennas can be broadly divided into two categories:

- (i) The Type-1 wideband antennas are produced by stimulated resonant modes that are either continuous or multiple. Spiral and log periodic antennas are two examples of these antennas [7].
- (ii) The Type-2 wideband antennas are produced by allowing emission from a limited wavelength area. Common

examples of Type-2 antennas include the Vivaldi antenna, the double-ridged horn, and the resistively terminated TEM horn [4], [5].

The increasing demand for multi-band and ultra-wideband systems has accelerated the need for antennas capable of maintaining stable performance over broad frequency ranges. Beyond the popular Vivaldi antenna, alternative designs such as spiral, sinuous, and log-periodic dipole arrays (LPDAs) offer diverse capabilities suited to various applications. While selecting the antenna type, all its advantages and disadvantages need to be analyzed.

(a) *Spiral Antenna:*

The spiral antennas are a type of wide-band antenna that generate circularly polarized waves and can be easily integrated with systems. They are a very suitable choice because of their frequency-independent nature. This is the class whose radiation pattern and impedance are practically independent of frequency for all frequencies above the minimum cut-off value [9]. The advantages include relatively easy impedance matching, better radiation efficiency, high spectral efficiency, consistent gain, and large bandwidth. The challenge of the spiral antennas is the feeding mechanism. Spiral antennas are inherently wideband, but the feed structure must maintain a consistent impedance (often 50 Ω) across a wide frequency range.

(b) *Sinuous Antenna:*

It can be seen as a combination of spiral and log-periodic antenna concepts, which result in a radiating element capable of producing wideband radiation in the broadside direction with polarization diversity. Non-self-complementary two-arm sinuous antenna and a self-complementary conical projection of the two-arms sinuous structure were introduced to realize unique multiband behaviour with alternating polarization handedness between adjacent bands [10]. Sinuous antennas have complex geometries that make fabrication and miniaturization challenging. They exhibit moderate gain and bidirectional radiation, which may require additional reflectors. Feeding issues in sinuous antennas are critical challenges that can significantly affect their broadband performance, polarization purity, and radiation patterns.

(c) *Log-Periodic Dipole Antenna (LPDA):*

LPDAs offer wide bandwidth with stable radiation patterns and consistent gain across frequencies. They support both linear polarization and good impedance matching over large frequency ranges [11]. Log-periodic dipole antenna printed on a PCB has many advantages, including low manufacturing cost, a high yield rate, and lightweight. The characteristic of this antenna changes with the logarithm of frequency, and expands the characteristic in one cycle to a wide channel. Normally, the log periodic antenna requires more elements for accepting and sending signals. Additionally, they are large and not suitable for systems with limited size or volume. It has a narrower beamwidth compared to other types of antennas. The assembly and maintenance of the log periodic antenna may be more complex and time-consuming due to the multiple elements and precise spacing required.

(d) *Vivaldi Antenna:*

The Vivaldi antenna is a type of end-fire traveling-wave antenna that is highly suitable for applications requiring a broad bandwidth and directional characteristics [12]. The core design of Vivaldi antennas includes an exponentially tapered planar slot, typically fed through electromagnetic

coupling by means of a microstrip-to-slot line transition. This antenna operates with a non-resonant radiation mechanism, where waves travel along the tapered slot, confined between the conductors.

In the design of a wideband antenna array tailored for seeker-based systems, several candidate antenna structures were evaluated, including spiral, sinuous, log-periodic dipole, and tapered slot (Vivaldi) antennas. Each configuration was assessed based on criteria such as bandwidth, gain, polarization stability, structural conformity, and suitability for compact integration. While spiral and sinuous antennas offer excellent bandwidth and polarization diversity, their bidirectional radiation and fabrication complexity pose limitations for conformal implementation. Similarly, log-periodic dipole arrays, though broadband, require larger apertures and are less planar. After a thorough comparative analysis, the Vivaldi antenna was selected as the optimal solution. Its inherent wideband characteristics, high directivity, low profile, and ability to conform to curved surfaces make it highly compatible with the compact, planar, and conformal constraints of seeker-based array applications.

III. ANTENNA DESIGN

On a 20 mil thick Rogers RT6035HTC substrate, with a loss tangent of 0.0023, a dielectric constant of 3.6, a Vivaldi-like antenna is designed in CST studio suite 2021. Considering the aforementioned design parameters into account, the basic structure of the proposed Vivaldi-like antenna using Co-planar waveguide (CPW) technology is demonstrated in Fig. 1. The considered antenna dimensions and their values are given in Table 1. The size of the antenna is 50×98 mm². Tapered slots are inserted to match the impedance at intermediate frequencies. Two small circular slots have been inserted between the feed arm and ground arm to cater to the radiation pattern at higher frequencies. The top layer indicates the exponential taper profile [7] which is defined by the opening rate, R and the two points P1 (x₁,y₁) and P2 (X₂,Y₂). The exponential taper profile is given by:

$$y = C_1 e^{Rx} + C_2 \text{ for } x_0 \leq x \leq x_1 \quad (1)$$

where R is the rate of opening and

$$C_1 = (y_1 - y_0) / (e^{Rx_1} - e^{Rx_0}) \quad (2)$$

$$C_2 = (e^{Rx_1} y_0 - e^{Rx_0} y_1) / (e^{Rx_1} - e^{Rx_0}) \quad (3)$$

Table 1: Considered specifications for the Vivaldi-like antenna.

Parameters	Dimensions(mm)
length (l)	98.09
width (w)	50.00
Height of substrate (h)	0.508
Length of microstrip feed line (Ml)	18.09
Width of microstrip feed line (Mw)	4.00

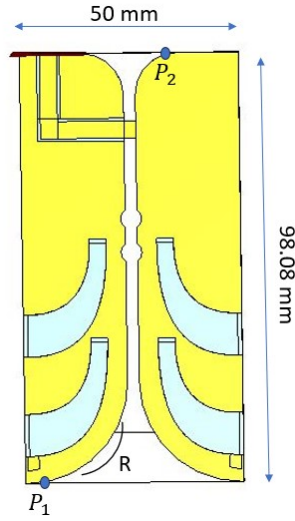


Fig. 1 Geometry of the proposed Vivaldi-like antenna.

IV. SIMULATION RESULTS

The proposed antenna is simulated in CST Microwave Studio. The reflection coefficient (S_{11}) is plotted in Fig. 2.

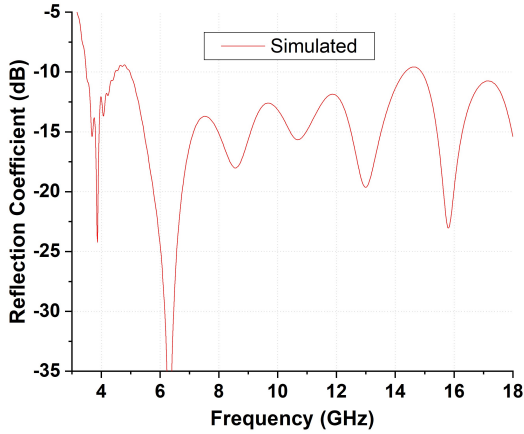


Fig. 2 Reflection coefficient (S_{11}) for a single Vivaldi-like structure.

The simulated S_{11} parameters, as illustrated in Fig. 2, demonstrate a good impedance match, hence a broad impedance bandwidth across the frequency spectrum. The simulated reflection coefficient results confirm wideband performance, with S_{11} remaining below -10 dB from 3 GHz to 18 GHz. This demonstrates the antenna's suitability for broadband applications requiring good return loss. The radiation characteristics of the proposed Vivaldi-like antennas, plotted with respect to spatial coordinates, are depicted in Fig. 3 at different frequencies.

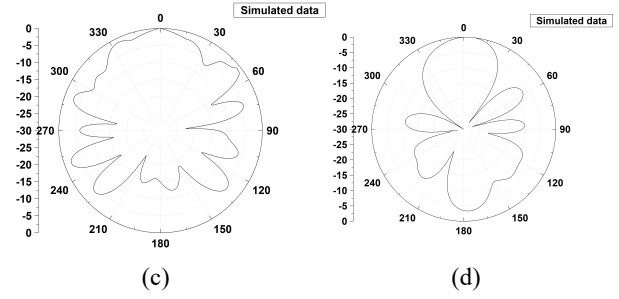
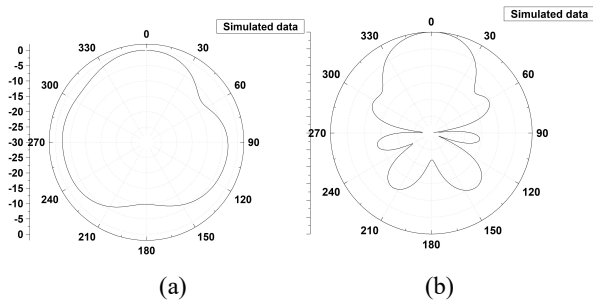


Fig. 3 Radiation patterns of the single-element Vivaldi-like antenna at (a) 5 GHz, (b) 9 GHz, (c) 13 GHz (d) 18 GHz.

The specific design parameters for each configuration, from a single element to a 4-element array, are provided beneath the corresponding figures. The 4-element array is designed after considering the Vivaldi-like antenna as the reference element, as illustrated in Fig. 4(a). The gap between the antenna elements is kept at 2 mm to protect it from shorting. The reflection coefficient is plotted in Fig. 4(b). The simulation results indicate that increasing the number of Vivaldi elements leads to an enhancement in antenna gain. Notably, the 4-element array configuration achieves the highest gain among all the designs evaluated. Radiation patterns are plotted in Fig. 5.

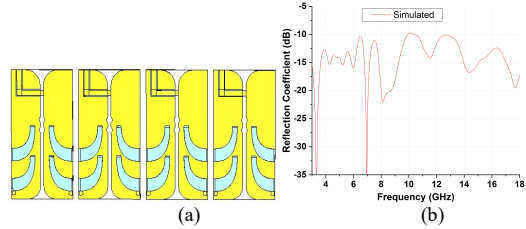


Fig. 4 (a) Four-element Vivaldi-like antenna array (b) Reflection coefficient (S_{11}) for an array of 4 elements

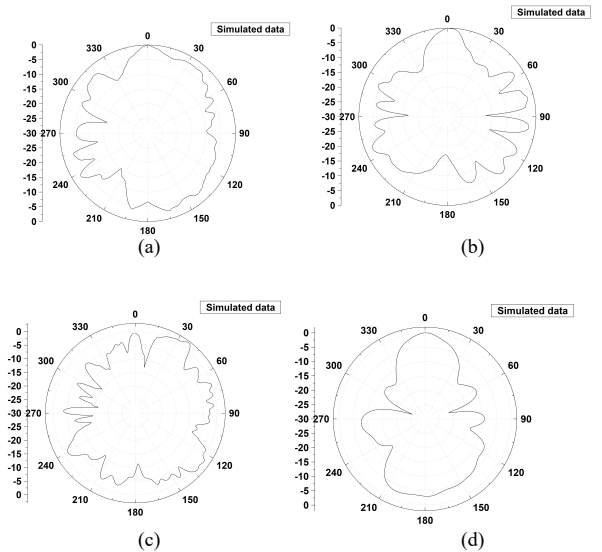


Fig. 5 Radiation patterns of the 4-element Vivaldi-like array at (a) 5 GHz, (b) 9 GHz, (c) 13 GHz, and (d) 18 GHz.

Radiation patterns of the array are distorted at the higher frequencies, which happens due to the increase in the gap between the elements at higher frequencies, which is more than $\lambda/2$. The challenge here is to maintain the side lobe

levels below the acceptable limit. It can be further improved by reducing the width of the single element.

V. CONFORMAL ARRAY

A 4×1 Vivaldi antenna structure has been conformally shaped to a cylinder of different diameters to see the impedance and radiation characteristics to be used for compact and high-performance wideband antenna systems. The configuration illustrated in Fig. 6 showcases an array of tapered slot antenna elements, integrated on a curved surface to suit the geometry of cylindrical platforms. The element layout and feeding network are carefully optimized to maintain consistent performance despite the surface curvature, enabling effective angle-of-arrival estimation and direction-finding capabilities. reflection coefficient (S_{11}) characteristics of the conformal Vivaldi-like antenna array were analyzed for three different cylindrical curvatures of diameters 23 cm, 27 cm, and 30 cm. The array maintains S_{11} below -10 dB across the operating spectrum as illustrated in Fig. 7. Furthermore, as the diameter increases, the resonant frequencies show a slight downward shift, which may be attributed to the impact of the larger curvature on the current distribution and effective electrical length.

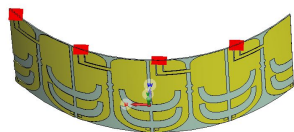


Fig. 6 Conformal Vivaldi antenna array

Overall, these results suggest that the designed antenna can be a good candidate for conformal shapes.

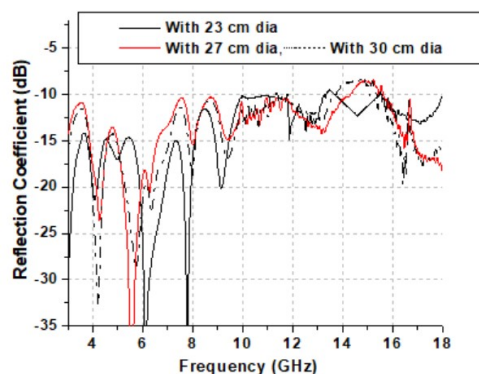


Fig. 7 Reflection coefficients (S_{11}) for Conformal Vivaldi-like antenna array with diameters of 23, 27, and 30 cm.

The radiation patterns were also observed for the conformal case. Radiation patterns show a similar behaviour as seen in the planar array case. The current design will undergo further refinement to enhance its performance and adaptability for real-world applications. Subsequent stages will include fabrication and experimental testing to validate the simulated results. Additionally, efforts will be made to extend the antenna's operating frequency range toward 2–18 GHz, aiming to achieve broader bandwidth coverage and improved functional capabilities. These advancements are expected to yield more efficient results, particularly in applications requiring high-resolution wideband performance.

VI. CONCLUSION

A compact, planar, and conformal Vivaldi-like antenna and its array have been presented to meet the requirements of wideband, compact, and platform-adaptable systems, particularly suited for Electronic Warfare Applications seeker. The design demonstrates the ability to retain the desirable characteristics of traditional Vivaldi antennas, such as broadband operation and directional radiation, while conforming to a curved surface for better mechanical integration. The simulation results indicate promising performance, validating the potential of this configuration for practical use. Future enhancements, including bandwidth extension and side lobe reduction, are expected to further improve the antenna's effectiveness and readiness for real-time deployment in advanced RF systems.

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